

In vitro investigations of the potential health benefits of Australian-grown faba beans (*Vicia faba* L.): chemopreventative capacity and inhibitory ...

The functional properties, including antioxidant and chemopreventative capacities as well as the inhibitory effects on angiotensin-converting enzyme (ACE), α -glucosidase and pancreatic lipase, of three Australian-grown faba bean genotypes (Nura, Rossa and TF(Ic*As)*483/13) were investigated using an array of in vitro assays. Chromatograms of on-line post column derivatisation assay coupled with HPLC revealed the existence of active phenolics (hump) in the coloured genotypes, which was lacking in the white-coloured breeding line, TF(Ic*As)*483/13. Roasting reduced the phenolic content, and diminished antioxidant activity by 10-40 % as measured by the reagent-based assays (diphenylpicrylhydrazyl, 2,2'-azino-bis(3-ethylbenzthiazoline-6-sulphonic acid) and oxygen radical absorbance capacity) in all genotypes. Cell culture-based antioxidant activity assay (cellular antioxidant activity) showed an increase of activity in the coloured genotypes after roasting. Faba bean extracts demonstrated cellular protection ability against H₂O₂-induced DNA damage (assessed using RAW264.7 cells), and inhibited the proliferation of all human cancer cell lines (BL13, AGS, Hep G2 and HT-29) evaluated. However, the effect of faba bean extracts on the non-transformed human cells (CCD-18Co) was negligible. Flow cytometric analyses showed that faba bean extracts successfully induced apoptosis of HL-60 (acute promyelocytic leukaemia) cells. The faba bean extracts also exhibited ACE, α -glucosidase and pancreatic lipase inhibitory activities. Overall, extracts from Nura (buff-coloured) and Rossa (red-coloured) were comparable, while TF(Ic*As)*483/13 (white-coloured) contained the lowest phenolic content and exhibited the least antioxidant and enzyme inhibition activities. These results are important to promote the utilisation of faba beans in human diets for various health benefits.